

(1116)

WORKAXIS — Intelligent Real-Time Helper Assignment Platform

Executive Summary

WorkAxis is a smart automation platform designed to connect users with verified house help professionals within 15 minutes. Traditional service-booking platforms often suffer from delays, unreliable helpers, and lack of trust. **WorkAxis** solves these problems through predictive automation, behavioral trust scoring, and intelligent assignment algorithms. Instead of assigning the nearest available helper, the system evaluates multiple reliability signals such as response speed, cancellation history, and GPS authenticity to select the most dependable professional. The platform integrates real-time tracking, fraud detection, dynamic pricing, and automated reassignment to ensure seamless service delivery. Built with a scalable architecture using PHP, MySQL, and AI APIs, **WorkAxis** demonstrates production-grade engineering and real-world applicability. The system is designed to transform on-demand service booking into a fast, secure, and trustworthy experience.

Problem Statement

"Create a system where users can book house help services and receive assistance within 15 minutes using smart automation."

Our Solution

WorkAxis introduces an intelligent matching engine that predicts helper availability, calculates reliability scores, and assigns the most suitable helper instantly. The platform analyzes real-time data such as helper location, past performance, and behavioral patterns. If a helper rejects a request, the system automatically reassigns another without user intervention. This ensures faster response times and improved success rates. The platform also includes real-time tracking, OTP verification, dynamic pricing, and fraud detection, creating a secure and efficient service ecosystem.

Technical Architecture

Frontend

The interface is built using HTML, TailwindCSS, and JavaScript. The design is lightweight, responsive, and optimized for fast loading. Real-time updates are handled through efficient polling mechanisms.

Backend

The backend is implemented in PHP with modular APIs handling booking, assignment, authentication, and payment processing. PHP was selected for its rapid development capability and seamless database integration.

Database

MySQL stores structured data including users, helpers, bookings, ratings, locations, and transactions. Indexed queries improve performance and reduce response time.

AI Integration

AI services process user requests, detect service type, analyze behavior patterns, and assist in fraud detection and demand prediction.

Data Flow

"User Request → Server API → Assignment Engine → Database → Scoring Algorithm → Helper Assigned → Tracking → Completion → Trust Update"

Security

"Security measures include prepared statements, encrypted passwords, CSRF protection, input validation, file upload restrictions, and rate limiting."

Key Features

- Smart auto-assignment engine
 - Behavioral Trust AI reliability score
 - Predictive helper availability engine
 - Live GPS tracking
 - Dynamic pricing system
 - Fraud detection intelligence
 - AI chatbot booking assistant
 - Custom material request system
 - Doorstep OTP verification
 - Fair assignment logic
 - Demand prediction analytics
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Complete Feature Set

1. Behavioural Trust AI (Primary USP)

- Reliability prediction score based on:
 - GPS honesty detection
 - cancellation history
 - response speed
 - job completion rate
 - rating consistency
- Prevents unreliable helpers from being assigned
- Displays trust level before booking confirmation

2. Core System Intelligence

- Smart auto-assignment engine selecting optimal helper using weighted scoring
- Automatic reassignment if helper rejects or times out
- Predictive helper availability engine estimating next free helper

- Reliability prediction scoring system
- Fair assignment algorithm for balanced workload distribution
- ETA prediction engine based on distance + behavior patterns

3. User Features

- Secure signup/login
- OTP verification
- Multi-language interface
- Voice booking request
- Text booking request
- Image upload for problem description
- Custom item request system
- Extra materials addition
- Booking history
- Favourite helper saving
- Wallet system
- Payment gateway integration
- Priority booking option
- Subscription plans
- Cashback system
- Live tracking map
- Booking status timeline
- Cancel booking
- Smart notifications

4. Helper Features

- Helper registration portal
- Aadhaar verification
- Photo verification
- Skill selection
- Experience display
- Rating display
- Availability toggle
- Earnings dashboard
- Withdrawal system
- Accept/reject booking
- Work history tracking
- Add custom services
- Add required materials list
- Set material pricing
- Profile promotion boost
- Area preference selection
- Performance analytics dashboard

5. AI & Automation Features

- AI chatbot booking assistant
- Natural language request interpretation
- Automatic skill detection
- Demand prediction system
- Surge prediction engine
- Fraud detection intelligence
- Fake booking detection

- Multi-account detection
- Review spam detection
- Location spoof detection
- Smart service suggestions

6. Location & Real-Time Tracking

- GPS tracking
- Nearest helper detection
- Live movement tracking
- Geo-fence arrival verification
- Location logs
- Service shortage detection zones
- Demand heatmap visualization

7. Pricing Intelligence Engine

Dynamic pricing formula:

Base + Distance + Demand + Rating + Urgency + Priority

Includes:

- surge pricing
- night charges
- holiday charges
- rating multiplier
- distance pricing
- extra material cost calculation

8. Material & Custom Service System

- Helper can list required materials
- User can add custom materials
- Extra charges auto calculated
- Bill upload option
- Approval system before purchase
- Items remain with customer after job

9. Trust & Verification Layer

- OTP verification
- Doorstep OTP confirmation
- Aadhaar verification
- Live selfie verification
- Device fingerprinting
- Fake review prevention
- Trust score system
- Background verification badge

10. Admin Panel

- Live bookings dashboard
- Revenue analytics
- Helper performance reports
- Complaint management
- Cancellation analytics
- Demand heatmap
- Manual assignment override

- User banning system
- Export reports

11. Security System

- SQL injection protection
- XSS protection
- CSRF tokens
- Secure file uploads
- Password hashing
- Role-based access control
- Suspicious activity detection
- Auto account lock
- Rate limiting

12. Optimization & Reliability Logic

- Queue system for high demand
- Retry assignment logic
- Backup helper auto-assign
- Load balancing logic
- API failure fallback system

13. Communication System

- In-app chat
- Voice message support
- Notification center
- Status timeline updates

- ETA countdown timer

14. Business & Growth Features

- Promo codes
- Referral rewards
- Loyalty benefits
- Subscription plans
- Helper advertisement promotion boosts

Market Comparison Analysis

To evaluate the effectiveness of **WorkAxis**, we compared it against existing service booking approaches currently used in the market. Most platforms focus primarily on listings and ratings rather than intelligent decision systems. The comparison below highlights how **WorkAxis** introduces measurable advantages through automation, predictive logic, and behavioural analysis.

Platform	Limitation	<u>WorkAxis</u> Advantage
Urban Company	Relies mainly on ratings and reviews	Behavioral Trust AI evaluates real reliability before assignment
Local Vendors	No real-time tracking or verification	Live GPS tracking + identity verification
Manual Booking	Slow and uncertain matching	Instant automated helper assignment within minutes

This comparison demonstrates that **WorkAxis** does not merely digitize booking — it fundamentally improves how service matching works by introducing predictive intelligence and trust-based automation. Instead of asking users to choose blindly, the platform calculates and assigns the most suitable helper automatically.

Real World Impact

WorkAxis is designed not only as a technical solution but as a system that delivers meaningful real-world benefits across users, workers, and communities.

User Safety & Trust

By implementing identity verification, behavioural trust scoring, and real-time tracking, WorkAxis ensures that users feel confident and secure about who is entering their home. This directly addresses one of the biggest concerns in service marketplaces — safety.

Employment Opportunities for Workers

The platform promotes fair job distribution using workload-balancing algorithms, ensuring that helpers receive consistent opportunities rather than work being concentrated among a few. This increases earning potential and platform fairness.

Time Efficiency

Traditional service booking may take hours or even days to arrange assistance. WorkAxis reduces this to minutes through automated assignment and predictive availability estimation. This time optimization benefits both users and service providers.

Digital Transformation of Local Services

WorkAxis bridges the gap between informal service sectors and digital platforms, enabling local workers to access structured employment opportunities while giving users modern, reliable service access.

Innovation & Differentiation

Unlike conventional platforms that rely on simple proximity matching, WorkAxis uses predictive intelligence and behavioral analytics. Its trust scoring system evaluates helper reliability before assignment rather than after service completion. Automated reassignment and workload balancing further enhance efficiency. This combination of predictive logic, automation, and security makes WorkAxis a next-generation service platform.

Development Challenges & Solutions

One major challenge was designing an assignment algorithm that balanced distance with reliability. This was solved by implementing a weighted scoring system. Another challenge involved real-time updates without complex infrastructure, which was addressed using optimized polling. Database performance issues were resolved through indexing and query

optimization. Fraud detection required pattern analysis, so rule-based anomaly detection logic was implemented.

Demo Flow

User logs in → submits request → system predicts availability → best helper assigned → live tracking starts → helper arrives and verifies OTP → service completed → payment processed → rating updates trust score.

Future Scope

Future versions may include mobile applications, voice-based booking, advanced demand heatmaps, multi-city deployment, and predictive scheduling systems. The modular architecture allows seamless scalability and feature expansion.

Team

Team Name: *MM Rangers (1116)*

1. Prathamesh Sawarkar (12509401)— Pitch and Developer
2. Moomin Bakshi (12518570)— Data Management & Resource Allocation
3. Malik Arhaan (12512702)— Developer and Presentation Designer
4. Rehu Talwar (12505449) — Developer

Conclusion

WorkAxis demonstrates how intelligent automation can transform on-demand service platforms. By combining predictive assignment, behavioral trust analysis, and real-time automation, the system significantly improves speed, reliability, and user confidence. It represents a scalable, secure, and commercially viable solution capable of redefining how users access household services.